LABEL: Literature Asymmetry between Efficacy and Limits in Drug Knowledge and Language Models

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Abstract

Every drug carries two kinds of fact: its efficacy, what it treats, and its limits, who must not take it. The limits protect patients, yet we find they are what prose writes down least. We measure, on the same facts, how often text names a drug's indications and contraindications and how often a language model knows them. Comparing each drug against itself on 574 drugs from PrimeKG, the odds that a benefit fact is named are 2.43 times those of a safety fact in Wikipedia, 2.77 times in PubMed abstracts, and 2.19 times in the two combined (95% CI 1.86–2.57). We call this the LABEL asymmetry (literature asymmetry between efficacy and limits). It disappears in the FDA drug label, scored with the same matcher on the same edges (odds ratio 1.05, 95% CI 0.88–1.25): safety facts are recorded where regulation requires them and left out of the prose. Models show the prose pattern. Across Qwen2.5 from 0.5B to 14B parameters, indication accuracy rises 18 points while contraindication accuracy stays within 7 points of chance at every size, and the gap reappears in four models from three families under harder distractors. Giving the model the drug's page helps safety facts about as much as benefit facts when the page names them ($+14.2$ against $+11.1$ points; difference -3.1, 95% CI -11.6 to $+5.1$), consistent with a documentation bottleneck rather than a reading failure. A control graph whose two relations both describe benefit shows no gap once each drug's number of edges is held fixed. Medication-safety questions should therefore be routed to the label and to structured drug databases rather than to text alone.

1 Introduction

Look up a medicine and the page will tell you what it treats. It is far less likely to tell you who must never take it. A drug's indications ("metformin treats type 2 diabetes") and its contraindications ("ciprofloxacin should not be given to people with myasthenia gravis") are both recorded in structured drug databases such as DrugCentral (Ursu et al. 2017). In free text, the two are not recorded equally.

This matters because language models learn from free text, and people now ask them about their medications. If the safety relation is the one text documents least, then the question with the highest cost of error is the one a model is least prepared to answer. It also locates the failure before the model. A model that misses a contraindication is not necessarily ignoring safety; the fact may never have been written where it could learn it. That changes the remedy. Better prompting and better retrieval over prose are unlikely to surface a fact that was never written down there, while a structured source can.

Two literatures touch this question without joining it. Pharmacy and informatics studies audit reference text, typically a small set of drugs compared against a textbook (Clauson et al. 2008; Kräenbring et al. 2014). Model benchmarks built from drug knowledge graphs score what models answer (Liu et al. 2026), but do not trace errors to the text. Neither measures text and models on the same facts, and neither controls for the obvious confound: famous drugs have long articles and many papers.

We join the two measurements on the same facts and compare each drug against itself (Figure 1). Our contributions are:

- A within-drug design over 574 PrimeKG drugs that compares the documentation of each drug's indications with its contraindications, so that article length, popularity, and research attention cancel.
- Three corpora with different authors and purposes: English Wikipedia, PubMed abstracts with a separate retrieval budget for each relation, and the FDA drug label. Prose shows the gap; the label, on the same edges, does not.
- A model-side probe on 1,800 facts across five sizes of one model family, a paired test of whether scale helps the two relations equally, and a replication under harder distractors in four models from three families.
- A retrieval test that gives the model the drug's page, separated by whether the page names the fact, which shows that retrieval helps safety facts fully when they are written down.
- A negative control on Hetionet, whose two drug–disease relations both describe benefit, and a routing design for clinical agents that follows from the measurements.

The finding, which we call the LABEL asymmetry, is simple: prose names a drug's efficacy about twice as often as its limits, the regulatory label does not, and models follow the prose.

2 Related Work

Drug information in reference text. Wikipedia is among the most consulted health resources on the web (Heilman

et al. 2011), and its drug articles have been audited several times. Clauson et al. (2008) compared Wikipedia with a professional drug compendium and found it less complete, with frequent omissions. Kräenbring et al. (2014) compared 100 drug articles with pharmacology textbooks and found high accuracy but incomplete coverage. These audits establish that coverage is uneven. They cover about a hundred teaching drugs, they grade text against a textbook rather than against a drug's full set of relations, and they do not connect coverage to what models learn.

Structured drug knowledge. Drug safety is recorded in resources built for that purpose. DrugCentral curates indications and contraindications from drug labels (Ursu et al. 2017), SIDER extracts side effects from package inserts (Kuhn et al. 2016), and openFDA exposes the full text of FDA structured product labels (Kass-Hout et al. 2016). Knowledge graphs such as PrimeKG and Hetionet integrate these sources with disease ontologies (Chandak, Huang, and Zitnik 2023; Himmelstein et al. 2017). These resources make it possible to ask, fact by fact, whether the same information also appears in free text.

Medical knowledge in language models. Large language models answer medical licensing questions well (Singhal et al. 2023, 2025), but benchmarks built from knowledge graphs show uneven knowledge across relation types. MH-GraphBench (Liu et al. 2026) probes models with questions generated from PrimeKG and reports that relation prediction is much harder than entity typing. These benchmarks describe what models get wrong; they do not trace the error to the text the models learned from.

Corpus frequency and model knowledge. A separate line of work links what models know to how often facts appear in pretraining data. Kandpal et al. (2023) show that question-answering accuracy rises with the number of documents relevant to a question, and Mallen et al. (2023) show that models fail on less popular entities, where retrieval helps most. LMEnt (Gottesman et al. 2026) annotates entity mentions in a pretraining corpus to trace knowledge to its sources. These studies measure entities or questions. Two facts about the same drug receive the same popularity score, so they cannot distinguish a drug whose benefits are written down from one whose risks are. Our measurement is per fact and compares relations within a drug.

Retrieval as a remedy. Retrieval-augmented generation (Lewis et al. 2020) is the standard answer to missing parametric knowledge: give the model a relevant document. In medicine, the MIRAGE benchmark (Xiong et al. 2024) shows that retrieval from PubMed, textbooks, and Wikipedia improves question answering, with gains that depend on the corpus and the retriever. These evaluations report accuracy per question set, not per relation, and do not ask whether the retrieved text contains safety facts at the rate it contains benefit facts. Work on sufficient context (Joren et al. 2025) shows that the benefit depends on whether the retrieved text actually contains the answer. Our results measure that condition for drug safety and show that it fails far more often for contraindications than for indications, in every prose corpus we tested. Bang et al. (2025) answer contraindication questions from a drug utilization review database, the routing our measurements support.

Source	Item	Count
PrimeKG	indication edges	9,388
	contraindication edges	30,675
	off-label use edges	2,568
Hetionet	treats / palliates edges	755 / 390
MONDO	parent links / diseases	17,801 / 11,788
Text sample	drugs	574
	edges per relation	1,722
FDA label sample	drugs with a label	472
	edges per relation	1,416
Probe sample	edges per relation	600
	model sizes (Qwen2.5)	5
Page test	ind. / contra. questions	450 / 406
	models (families)	4 (3)

Table 1: Data used in this study.

3 Data

Knowledge graphs. PrimeKG (Chandak, Huang, and Zitnik 2023) integrates 20 biomedical resources. Its drug–disease relations come from DrugCentral and comprise 9,388 indication edges, 30,675 contraindication edges, and 2,568 off-label use edges. Diseases are identified by MONDO (Vasilevsky et al. 2022), which we rebuild as a directed child-to-parent table of 17,801 edges over 11,788 diseases. As a control we use Hetionet (Himmelstein et al. 2017), built by a different group from different sources, with 755 *treats* and 390 *palliates* edges.

Corpora. For each drug we fetch its English Wikipedia article. For the second corpus we retrieve up to 40 relevance-ranked PubMed abstracts per drug through the NCBI E-utilities (Sayers 2009), with the budget split evenly between two queries on the drug's name: one framed for therapeutic use and one for adverse effects and contraindications (20 abstracts each). The third corpus is the FDA structured product label, retrieved through the openFDA drug label service (Kass-Hout et al. 2016). Each drug is searched by generic name, then by substance name, then by brand name, and up to five labels per drug are pooled, so that a drug sold by several manufacturers is covered by all of them. We keep three texts per drug: the Indications and Usage section, the Contraindications section, and the whole label (every text section, including Boxed Warning, Warnings and Precautions, and Use in Specific Populations). A label is found for 472 of the 574 drugs (82.2%).

Samples. The text comparison draws three indication and three contraindication edges from each of the 595 drugs that carry at least three of each. Of these, 574 drugs have an English Wikipedia article long enough to measure, which gives 1,722 edges per relation; the 472 drugs with a label give 1,416. The model probe uses a separate sample of 600 edges per relation for indication, contraindication, and off-label use (1,800 edges). Table 1 summarizes the samples.

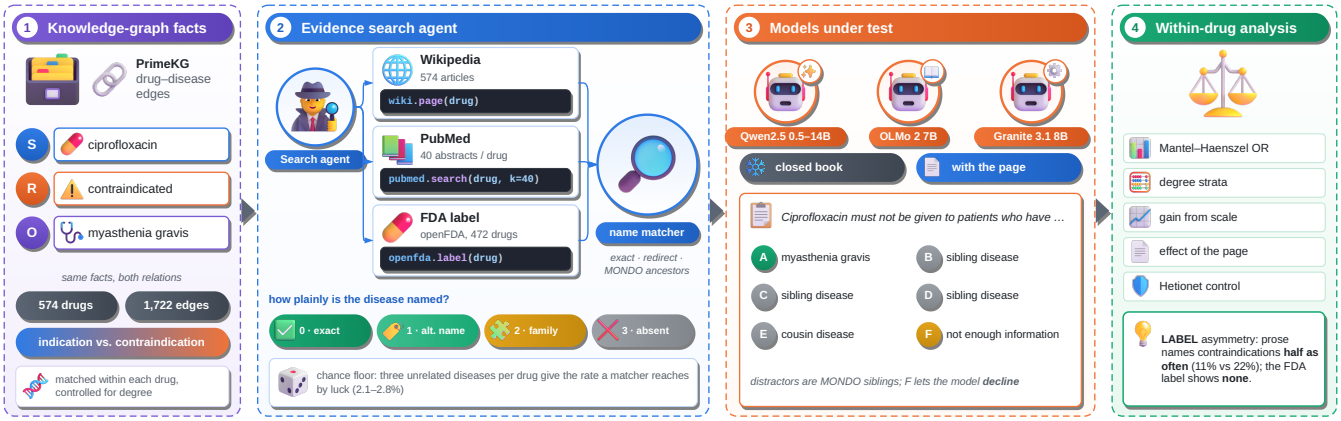


Figure 1: Study design as an agent pipeline. (1) PrimeKG supplies both relations for the same drugs; (2) a search agent retrieves Wikipedia, PubMed and FDA label text, and a name matcher scores how plainly each names the disease; (3) the models under test answer the same facts closed book and with the page; (4) the two relations are compared within each drug, which yields the LABEL asymmetry.

Level	Text contains	Example
0	exact name	“type 1 diabetes mellitus”
1	an alternative name	“juvenile diabetes”
2	only the family	“diabetes”
3	nothing	—

Table 2: Levels at which text can name a disease. Levels 0 and 1 count as named.

4 Method

4.1 Does the text name the fact?

For each edge, we check whether the drug’s text names the disease (Table 2): by its exact name, by an alternative name, or only by an ancestor category in MONDO. Matching is case-insensitive and respects word boundaries, so “asthma” does not match “asthmatic.” Alternative names come from Wikipedia redirects to the disease’s article, which include synonyms, lay terms, and abbreviations. Ancestors are taken up to two levels above the disease. Phrases shorter than four characters are not matched. The same matcher, with the same names, is applied to all three corpora.

We count a fact as *named* when the text contains the exact name or an alternative name. A category match alone is not counted, because “diabetes” does not say which diabetes.

String matches can occur by coincidence. To measure this, each edge is paired with three diseases from the same relation that the drug has no relation to, and the same matcher is run on them. In Wikipedia, on a draw of 600 edges per relation, the exact name of an unrelated disease appears in 1.22%, 0.72%, and 0.78% of indication, off-label, and contraindication cases. Labels are longer, and the floor on the whole label is 2.10% for indications and 2.82% for contraindications.

4.2 Comparing each drug against itself

A drug’s indications and contraindications share one article, one body of literature, and one label. We build a 2×2 table per drug (relation by named) and pool the tables with the Mantel–Haenszel odds ratio (Mantel and Haenszel 1959),

$$\mathrm{OR}_{\mathrm{MH}} = \frac{\sum_d a_d\, b'_d / n_d}{\sum_d b_d\, a'_d / n_d}, \tag{1}$$

where for drug d, a_d and b_d are its named and unnamed indications, a'_d and b'_d its named and unnamed contraindications, and n_d its total number of sampled edges. Its 95% confidence interval uses the test-based standard error $1/\sqrt{\sum_d V_d}$, where V_d is the hypergeometric variance of drug d’s table. Because each drug is its own stratum, article length, popularity, and research attention cancel. An odds ratio above 1 means indications are named more often than contraindications of the same drug.

Drugs differ in how many edges they carry, and a drug with many contraindications may name each one less often. To test this, we fit a logistic model of whether a fact is named on its relation and the logarithm of the drug’s edge count in that relation, and we compute the gap within strata of that edge count: 3 to 5, 6 to 12, and 13 or more.

4.3 Does the model know the fact?

Each probe edge becomes a closed-book multiple-choice question with five options: the true disease and four diseases that are not among the drug’s partners for that relation in the sample. No passage or retrieval is given. Each option is scored by its mean token log-probability as the completion of a relation-specific prompt, and the highest-scoring option is the model’s answer. Chance accuracy is 20%. The prompts are in the supplement. We probe Qwen2.5-Instruct at 0.5B, 1.5B, 3B, 7B, and 14B parameters (Qwen Team 2024), all in 32-bit precision on the same questions.

To test whether scale helps the two relations equally, we use the fact that all models answer the same edges. For each edge we compute the gain (larger model correct minus smaller model correct) and compare the mean gain between indications and contraindications. For each model size we also report the indication-minus-contraindication gap. Confidence

Corpus	Ind.	Contra.	OR [95% CI]	p
Wikipedia	20.33	9.81	2.43 [2.01, 2.94]	2×10^{-18}
PubMed	26.31	11.44	2.77 [2.33, 3.28]	5×10^{-29}
Both	29.04	15.91	2.19 [1.86, 2.57]	2×10^{-20}

Table 3: Percent of facts named in prose and the within-drug odds ratio, indication against contraindication. 574 drugs, 1,722 edges per relation.

intervals come from bootstrap resamples of drugs, so that edges of the same drug move together.

4.4 Harder distractors and the drug's page

Random distractors can make some questions easy by elimination. As a robustness check, we use a second question set whose wrong options are drawn from the true disease's MONDO neighborhood: its siblings first, then its cousins, and random diseases only when those run out. A sixth option reads "Not enough information." It covers 450 indication and 406 contraindication edges, in three groups of up to 150 by what the drug's page says about the disease: it names it, it names only its family, or it is silent. The questions are asked to four instruction-tuned models from three families: Qwen2.5 0.5B and 7B, OLMo 2 7B (Team OLMo et al. 2024), and Granite 3.1 8B (IBM Granite Team 2024).

Each question is asked twice: closed book, and with a passage from the drug's Wikipedia article and abstracts placed before the question. The passage is the one a retriever centered on the drug would return, cleaned of markup. The change in accuracy between the two conditions, within each group, measures what retrieval adds when the page names the fact and when it does not. Because the two relations have different mixes of groups, we report gaps within each group, combined with weights $n_1 n_2/(n_1 + n_2)$.

4.5 Negative control and statistics

If the measurement itself favored one relation over another, it would produce a gap between any two relations. Hetionet's *treats* and *palliates* both describe benefit, so a sound measurement should show little or no gap between them. We run the same text measurement on Hetionet, using the Disease Ontology for its hierarchy, and apply the same within-drug and edge-count tests.

Percentages carry 95% Wilson intervals. Within-drug comparisons use the Cochran–Mantel–Haenszel test with each drug as a stratum. Changes with the page are paired by question, with 95% intervals from 5,000 bootstrap resamples of questions; pooled estimates average each question over the three 7–8B models. The supplementary material gives the matching rules, prompts, retrieval queries, and full result tables.

5 Results

5.1 Prose names benefit facts twice as often

Table 3 and Figure 2 give the main result. In Wikipedia, 20.33% of indications are named in the drug's article against

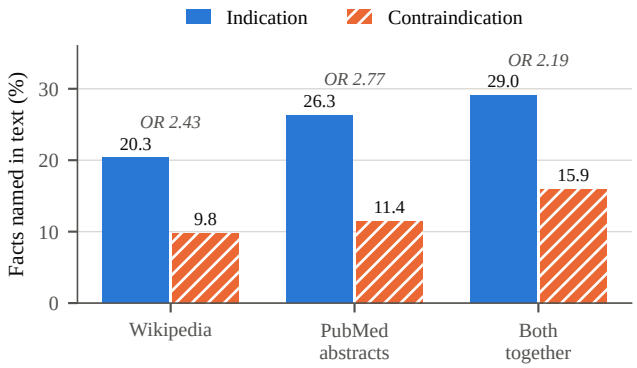


Figure 2: Share of facts whose text names the disease, for the same 574 drugs. Labels give the within-drug Mantel–Haenszel odds ratio.

9.81% of contraindications. Within the same drug, the odds of a benefit fact being named are 2.43 times those of a safety fact. PubMed abstracts give a gap of the same size, 2.77, so the asymmetry is not a property of the encyclopedia. The union of both corpora is the most generous prose baseline available: even there, the odds ratio is 2.19 (95% CI 1.86–2.57).

Searching harder does not close the gap. A natural objection is that general retrieval favors abstracts about what a drug treats. The PubMed corpus therefore gives the adverse-effect and contraindication query its own half of the budget. Over the same 1,722 contraindication edges, coverage was 10.22% with a single relevance-ranked query of 40 abstracts and 11.44% with the split budget, while indication coverage rose from 23.34% to 26.31% under the same change. The within-drug odds ratio is 2.77 under both retrieval schemes.

The gap concentrates in drugs with few facts. Figure 4 shows the gap by the drug's number of edges in the relation. It is 13.94 points for drugs with 3 to 5 edges in Wikipedia, 6.04 for 6 to 12, and 1.64 for 13 or more. The combined corpora show the same decline (11.81, 6.90, 2.23). In the logistic model, the relation term remains significant with the edge count held fixed ($p = 2.4 \times 10^{-4}$ in Wikipedia, 1.4×10^{-5} in PubMed, 9.7×10^{-4} combined). The gap is largest for the typical drug, which carries a handful of each.

Some missing evidence is coarser, not absent. Relaxing the match shows where the rest of the prose's information sits. On the draw of 600 edges per relation in Wikipedia, adding synonyms and common names to the exact name raises coverage by 1.39 points on average across relations. Adding the disease's MONDO parent raises it by a further 5.50 points, and adding ancestors up to two levels by 8.43. More than four fifths of the evidence gained by relaxing the match is written at a coarser level than the graph records, for example "liver disease" for a fact about cirrhosis. Such text is true and relevant, but it does not identify the fact, so we do not count it as named.

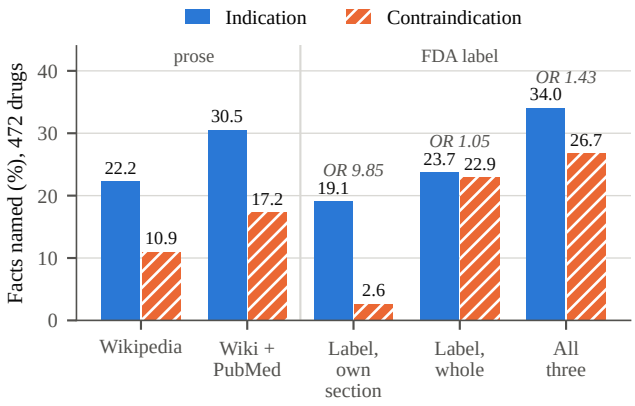


Figure 3: Share of facts named, on the 472 drugs with an FDA label. "Own section" is Indications and Usage for indications and Contraindications for contraindications. Labels give the within-drug odds ratio.

Text	Ind.	Con.	OR [95% CI]	Deg.
Wikipedia	22.25	10.95		
Wiki + PubMed	30.51	17.23		
Label, own section	19.07	2.61	9.85 [7.75, 12.51]	ind.
Label, whole	23.73	22.88	1.05 [0.88, 1.25]	contra.
All three	34.04	26.69	1.43 [1.21, 1.68]	none

Table 4: Percent of facts named on the same 1,416 edges per relation (472 drugs). "Deg." is the relation favored once the drug's edge count is held fixed in the logistic model: *ind.* and *contra.* significant at $p < 0.01$, *none* $p = 0.86$. Within-drug p: 10^{-45}, 0.62, and 2×10^{-5}.

5.2 The FDA label shows no detectable gap

If the gap came from the facts themselves, for example because some contraindications are too rare to write about, it would appear in every source. It does not. Scored with the same matcher on the same 1,416 edges per relation, the whole FDA label names 23.73% of indications and 22.88% of contraindications (Figure 3, Table 4). The within-drug odds ratio is 1.05 (95% CI 0.88–1.25, $p = 0.62$). With the drug's edge count held fixed, the label names contraindications slightly more often than indications (logistic coefficient $+0.30$, $p = 0.005$), and in no degree stratum does it favor indications (Figure 4). The difference in chance floors between the two relations (2.82% against 2.10%) is too small to account for this.

Where in the label. The label's own Contraindications section names only 2.61% of PrimeKG's contraindications, and for 9.5% of contraindication edges the section is empty. The harm facts sit elsewhere in the label: in the Boxed Warning, in Warnings and Precautions, and in Use in Specific Populations. A retrieval system that reads only the Contraindications section of a label misses most of them.

Adding the label to prose. Combining all three corpora raises coverage to 34.04% and 26.69% and lowers the odds ratio to 1.43; with edge counts held fixed the relation term is no longer significant ($p = 0.86$). Of the 1,172 contraindications that neither Wikipedia nor PubMed names, the label names 11.4%. The asymmetry is therefore a property of prose, not of the facts: the record that regulation requires documents harm as often as benefit, and the text that general-purpose models read does not.

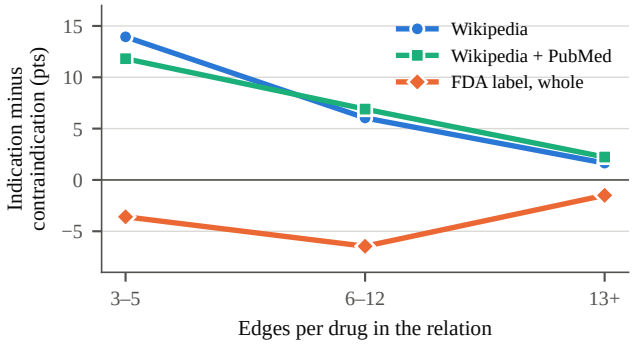


Figure 4: Gap between indication and contraindication coverage, by the drug's number of edges in the relation. Prose favors indications in every stratum; the whole FDA label does not.

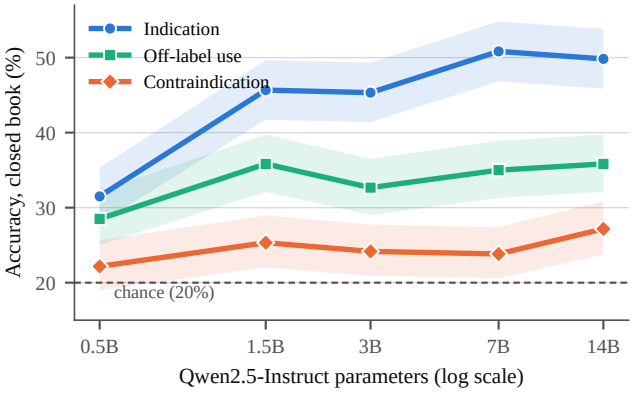


Figure 5: Closed-book accuracy on 600 edges per relation, five options, across five sizes of Qwen2.5-Instruct. Bands are 95% Wilson intervals.

5.3 Scale buys benefits, not harms

Figure 5 and Table 5 show the model side. Across a 28-fold range in parameters, indication accuracy rises from 31.50% to about 50%, flattening from 7B. Contraindication accuracy stays between 22.17% and 27.17% at every size, within 7 points of chance. The gap opens by 1.5B and stays between 20 and 27 points through 14B. The paired test between 0.5B and 7B confirms that scale helps the two relations unequally: the gain on indications (19.33 points) exceeds the gain on contraindications (1.67 points) by 17.67 points (95% CI 12.68–22.70, $p = 4.8 \times 10^{-12}$, 709 drugs resampled). From 0.5B to 14B, the gains are 18.33 points for indications, 7.33 for off-label use, and 5.00 for contraindications.

The ordering of accuracy follows the ordering of prose

Qwen2.5	Ind.	Off-label	Contra.	Gap [95% CI]
0.5B	31.50	28.50	22.17	+9.3 [4.0, 14.9]
1.5B	45.67	35.83	25.33	+20.3 [15.2, 26.2]
3B	45.33	32.67	24.17	+21.2 [16.1, 26.9]
7B	50.83	35.00	23.83	+27.0 [21.6, 32.4]
14B	49.83	35.83	27.17	+22.7 [17.1, 28.3]
Wikipedia names it	17.17	10.50	5.50	
Chance floor	1.22	0.72	0.78	

Table 5: Closed-book accuracy (%) on the probe sample, chance 20%. "Gap" is indication minus contraindication, with a 95% bootstrap interval over drugs. The last two rows give the exact-name match in Wikipedia for the same edges and its chance floor.

Model	Ind.	Contra.	Gap	Within	p
Qwen2.5 0.5B	36.0	28.1	+7.9	+8.2	0.009
Qwen2.5 7B	56.0	25.1	+30.9	+31.3	$< 10^{-15}$
OLMo 2 7B	56.0	33.0	+23.0	+23.6	2×10^{-13}
Granite 3.1 8B	58.7	12.8	+45.9	+46.3	$< 10^{-15}$

Table 6: Closed-book accuracy (%) with distractors from the answer's MONDO neighborhood plus a "Not enough information" option; 450 indication and 406 contraindication questions. "Within" is the difference within groups of questions whose page names the answer, its family, or nothing, which sets the p-value.

coverage across all three relations: indications are named most often and answered best, contraindications are named least often and answered worst, and off-label use sits between them on both measures. This is the pattern expected when the training signal for a relation is thin, although our data do not measure the models' training corpora directly.

The gap holds under harder distractors. Table 6 repeats the comparison with wrong options drawn from the true disease's own neighborhood in MONDO. The gap appears in all four models and three model families, from 7.9 points in Qwen2.5 0.5B to 45.9 in Granite 3.1 8B, and it holds within each naming group.

Two ways of not knowing. The "Not enough information" option separates two failure styles (Figure 6, Table 7). Qwen2.5 0.5B and OLMo 2 7B rarely decline and answer most contraindication questions wrongly (71.4% and 64.0% wrong). Qwen2.5 7B and Granite 3.1 8B decline most of them (50.7% and 67.5%) while declining indications far less (22.0% and 12.7%). In every model, precision on the contraindications it does answer is lower than on indications, by 9 to 28 points. Neither style supplies the fact: one guesses, the other abstains.

5.4 Retrieval helps safety facts when they are written

If models failed on contraindications because they could not use safety information, giving them the page would help indications more. It does not. When the page names the disease, the passage raises accuracy by 11.1 points on indications and 14.2 points on contraindications, pooled over the three 7–8B models (Figure 7, Table 8). The difference, -3.1 points (95% CI -11.6 to $+5.1$), is not significant, and the pattern holds in each model.

When the page is silent, accuracy falls with the page for both relations, by 11.3 and 8.9 points. The loss comes from declines: with a silent page, Qwen2.5 7B answers "Not enough information" on 94.0% of contraindication questions against 56.0% closed book, and Granite on 92.7% against 71.3%. That is the right response to a page that does not contain the answer. Pooled over the three 7–8B models (table in the supplement), the shift is the same. Declines rise most with a silent page (by 32.9 points on contraindications and 26.0 on indications), less with a page that names only the family (12.3 and 16.9), and barely with a page that names the answer (5.4 and -0.2).

Retrieval therefore helps safety facts about as much as benefit facts; we detect no difference. What differs is how often there is anything to retrieve: in prose, 17% of contraindications against 30% of indications on the label sample.

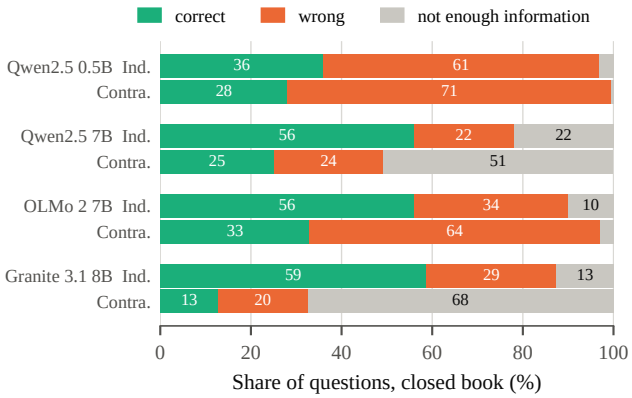


Figure 6: Outcome of each closed-book question with harder distractors. On contraindications, two models mostly guess wrong and two mostly decline.

	Indication			Contraindication		
Model	wrong	decl.	prec.	wrong	decl.	prec.
Qwen2.5 0.5B	60.7	3.3	37.2	71.4	0.5	28.2
Qwen2.5 7B	22.0	22.0	71.8	24.1	50.7	51.0
OLMo 2 7B	34.0	10.0	62.2	64.0	3.0	34.0
Granite 3.1 8B	28.7	12.7	67.2	19.7	67.5	39.4

Table 7: How models fail, closed book, harder distractors (%). "Decl." is the share answering "Not enough information"; "prec." is accuracy among questions the model answered.

5.5 The control graph shows no gap

On Hetionet, Wikipedia names the disease for 57.33% of *treats* edges and 56.10% of *palliates* edges (odds ratio 1.05,

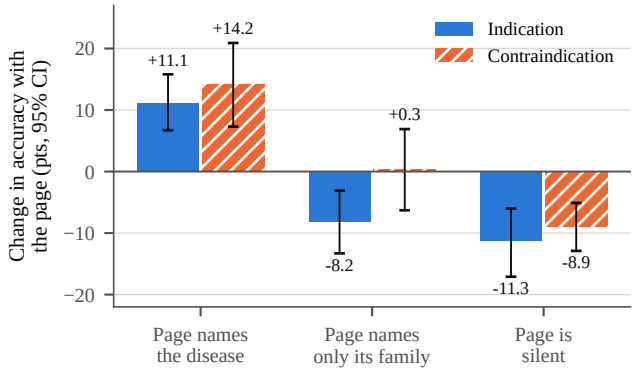


Figure 7: Change in accuracy when the drug's page is placed in the prompt, pooled over the three 7–8B models, by what the page says about the answer.

	Names it		Family only		Silent	
Model	Ind.	Con.	Ind.	Con.	Ind.	Con.
Qwen2.5 0.5B	+39.3	+40.7	+10.7	+3.8	+7.3	−6.0
Qwen2.5 7B	+6.7	+10.7	−12.0	−7.5	−16.7	−14.0
OLMo 2 7B	+12.7	+12.7	−2.7	0.0	−1.3	−5.3
Granite 3.1 8B	+14.0	+19.3	−10.0	+8.5	−16.0	−7.3
Pooled 7–8B	+11.1	+14.2	−8.2	+0.3	−11.3	−8.9

Table 8: Change in accuracy (points) from closed book to the drug's page, by what the page says about the answer. 150 questions per cell, except 106 family-only contraindication questions.

$p = 0.70$), against a within-drug odds ratio of 2.43 between indications and contraindications in PrimeKG on the same corpus. Within the 51 drugs that carry both Hetionet relations, *treats* edges are named more often (odds ratio 2.48, 95% CI 1.29–4.75), but this difference disappears once the drug's edge count is held fixed: the relation term in the logistic model has $p = 0.16$, and the degree strata show no consistent direction. The PrimeKG gap survives the same control. Together with the FDA label, which shows no gap under the same matcher, the control shows that the measurement does not produce a gap where the sources do not have one.

5.6 Summary of checks

Table 9 lists the alternative explanations we tested and the result of each. Each explanation that would make the prose gap an artifact of the drugs, the corpora, the retrieval, the matcher, or the models is addressed by at least one test, and the FDA label argues against an explanation in terms of the facts themselves.

6 Discussion

Safety is the blind spot of prose. Across two prose corpora that share no editorial process, the facts that matter most for harm are the ones least often written down. The label, which regulation requires to carry them, shows no such gap. One explanation is that prose is organized around what a drug is for: the encyclopedia for a general reader, the literature around efficacy. Even abstracts retrieved for toxicity rarely name the specific condition that forbids a drug.

Explanation	Test	Result
Famous drugs dominate	Within-drug MH odds ratio	2.19–2.77 in prose
Many contraindications per drug	Logistic model, degree strata	Gap holds, $p \leq 10^{-3}$
One editorial process	Wikipedia and PubMed separately	2.43 and 2.77
Retrieval favors efficacy	Split PubMed budget	OR 2.77 both ways
Matcher favors a relation	Hetionet, two benefit relations	No gap, $p = 0.16$
Coincidental matches	Floor on unrelated diseases	$\leq$ 1.2% prose, $\leq$ 2.8% label
Facts are too rare to write	Whole FDA label	OR 1.05, $p = 0.62$
Small models only	Five sizes, 0.5B–14B	Contra. within 7 pts of chance
Easy distractors	MONDO siblings, 4 models	Gap 7.9–45.9 pts
Models cannot use safety text	Page in the prompt	+14.2 vs. +11.1 pts

Table 9: Alternative explanations for the gap and the tests that address them.

Scale did not fix it. If contraindication errors came from limited capacity, scale would reduce them. Across five sizes, the gain goes almost entirely to indications and contraindication accuracy stays near chance. Larger models also change how they fail rather than whether they know: the 7–8B models in Table 7 either guess or decline on most safety questions.

What this means for clinical agents. The special theme of this year's workshop is agentic biomedical and clinical workflows. An agent that answers a medication question by reading prose, or by consulting its own weights, is exposed to the asymmetry measured here. Figure 8 shows a design motivated by the measurements; it is a proposal, not an evaluated clinical system. A controller first classifies the question by relation. Benefit questions can be answered from prose with an evidence check, since prose names 29–31% of indications and the page helps when it does. Safety questions go to a structured source such as DrugCentral and to the whole FDA label, not only its Contraindications section, which names 2.6% of PrimeKG's contraindications. An evidence check then decides between answering with the matched sentence and declining with the name of the source that was silent. The per-fact matcher used here is cheap enough to serve as that check.

What this means for retrieval design. Three practical rules follow. First, the corpus matters by relation: adding the label to prose closes most of the safety gap (odds ratio 2.19 to 1.43), while adding more prose of the same kind does not (2.43 to 2.19). Second, the section matters: a label retriever that indexes only the Contraindications section sees 2.6% of

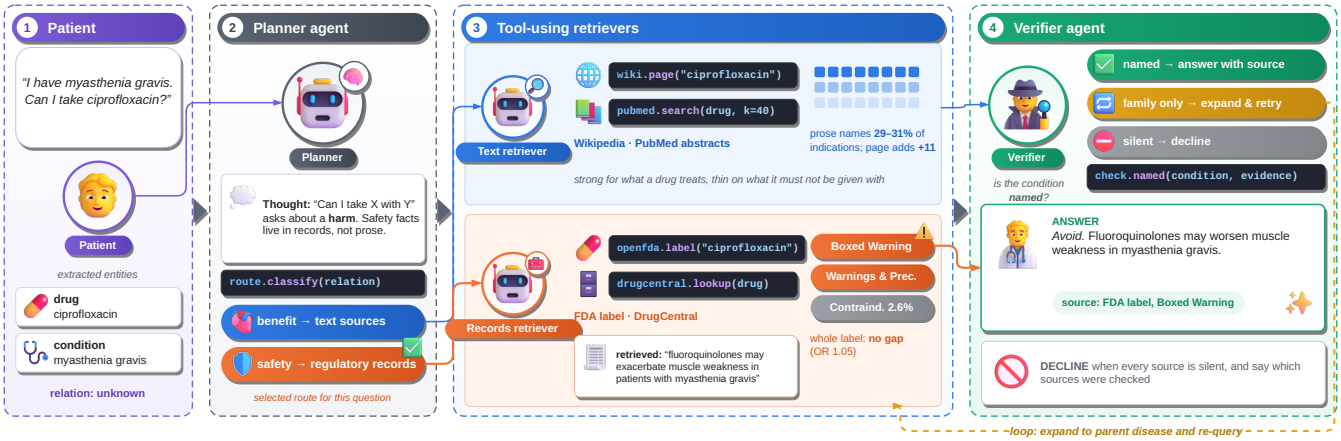


Figure 8: A relation-aware routing design for clinical agents that follows from the measurements. A controller classifies the question as benefit or safety and chooses the retrieval route; a checker decides between answering with a cited source, expanding a family-level mention, and declining with the name of the silent source. The example answer paraphrases the Boxed Warning of the ciprofloxacin label.

the safety facts that the whole label names 22.9% of. Third, much of what prose does say about a disease is said at the level of its family, and such a passage is relevant without being sufficient; a system should treat it as a reason to look further, not as an answer.

Off-label use sits in between. Off-label uses are benefit facts that a drug's label does not approve. They are named in prose less often than indications and more often than contraindications, and model accuracy and its gain from scale fall in the same middle position. Across the three relations, the ordering of model accuracy follows how often text names them, not whether they describe benefit or harm.

What this means for evaluation. Medical benchmarks usually report one accuracy per model. Our results show that the average hides a relation that sits near chance and does not improve with scale. Benchmarks built from knowledge graphs should report accuracy per relation, and in particular for contraindications, so that gains on benefit facts are not read as gains in safety.

An example. A patient with myasthenia gravis asks whether an antibiotic is safe for them. PrimeKG records that ciprofloxacin is contraindicated in this disease. Across our sample, a contraindication of this kind is named in its drug's Wikipedia article about one time in ten, a 7B model picks it out of five options about one time in four, and the label, read in full, names safety facts as often as benefit facts.

7 Limitations

Corpus to model. The link from corpus to model is correlational: we measure public corpora, not the models' training data. Two prose corpora with the asymmetry, a label without it, and models that follow the prose make the link plausible, but the causal chain is not demonstrated. PubMed figures come from abstracts and are a lower bound.

Matching. String matching misses paraphrases and can match a name used in an unrelated sense; the floor bounds the second error at under 3%. A name match also does not show that the text asserts the relation, which a blinded audit of matched sentences would measure. Both errors apply to both relations on the same text.

Models and questions. The scale curve uses one model family; the distractor and page tests add two more at one size each. Multiple choice measures recognition rather than recall.

Graph labels. Even the whole label names only about a quarter of PrimeKG's sampled contraindications, so many edges are recorded more specifically than any text states them. A clinical review of unnamed contraindications would separate specificity from omission.

8 Ethical Considerations

The study uses only public graphs, text and drug labels, and no patient data. Its results are not guidance on any individual drug: a fact's absence from prose does not make it uncertain, and the contraindications studied here remain authoritative in the sources that record them. Code that rebuilds every number is released with the paper.

9 Conclusion

The LABEL asymmetry is specific to prose: text names a drug's efficacy about twice as often as its limits, and the FDA label does not. Medication-safety questions should be routed to the label.

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